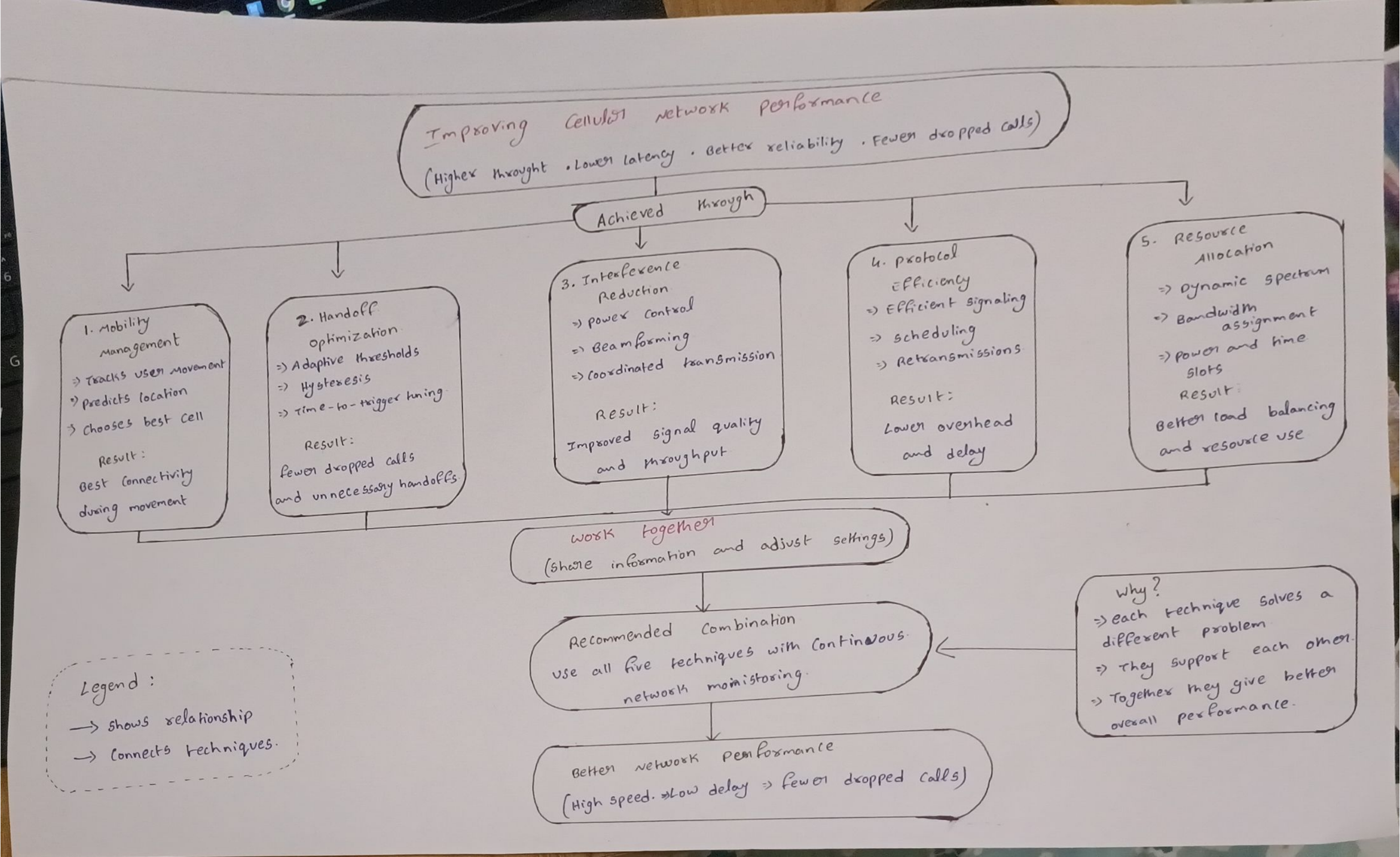

Improving Cellular Network Performance
(Higher throught • Lower latency • Better reliability • Fewer dropped calls)
Achieved through
1. Mobility Management
=> Tracks user movement
=> Predicts location
=> Chooses best cell
Result:
Best connectivity during movement
2. Handoff Optimization
=> Adaptive thresholds
=> Hysteresis
=> Time-to-trigger tuning
Result:
Fewer dropped calls and unnecessary handoffs
3. Interference Reduction
=> power control
=> Beamforming
=> Coordinated transmission
Result:
Improved signal quality and throughput
4. Protocol Efficiency
=> Efficient signaling
=> scheduling
=> Retransmissions
Result:
Lower overhead and delay
5. Resource Allocation
=> Dynamic spectrum
=> Bandwidth assignment
=> Power and time slots
Result:
Better load balancing and resource use
work together
(share information and adjust settings)
Recommended Combination
use all five techniques with Continuous network monitoring
Better network Performance
(High speed. => Low delay => Fewer dropped calls)
Why?
=> each technique solves a different problem
=> They support each other
=> Together they give better overall performance.
Legend:
→ shows relationship
→ Connects techniques.